AI ML Internship Project

Presented By – The Hunters

Title - An AI solution to predict crop yields and optimize farming practices based on weather patterns, soil health, and crop management techniques. The goal is to increase agricultural productivity and reduce hunger in rural communities.

Subtitle - AI-Based Crop Yield Prediction System for Smart Farming

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Introduction

- **Agriculture Overview:** Farming is the backbone of rural India, yet crop yields often suffer due to changing weather and limited planning tools.

- **Current Challenge:** Farmers lack accurate, real-time predictions to make informed decisions about crop production.

- **Proposed Solution:** A simple AI-powered web app that predicts crop yield using inputs like rainfall, temperature, and pesticide usage to support smart farming.

Problem Statement

Objective: To predict crop yield using AI models through a user-friendly web interface based on real-time agro-environmental data.

Specific Goals:

- Predict crop yield using AI and environmental data.
- Assist farmers in making informed farming decisions.
- Increase agricultural productivity through data insights.
- Promote sustainable farming practices with technology.
- Provide a user-friendly web app for easy crop prediction

Proposed Solution

- Develop a web-based application using machine learning to predict crop yield.
- Users input key factors like crop type, rainfall, temperature, pesticides, etc.
- The model processes the data and provides an estimated crop yield (kg/ha).
- Built using Streamlit for a simple and interactive user interface. Aims to support farmers and planners in making accurate, data-driven decisions for better agricultural output.

Technology Stack

Programming Language

PYTHON Chosen for its simplicity and rich ecosystem of libraries in data science and machine learning.

Data Collection & Storage

CSV AND EXCEL FILES Used for storing small to medium-sized datasets efficiently.

Machine Learning & Data Processing Libraries

SCIKIT-LEARN For implementing classic ML algorithms (e.g., Linear Regression, Random Forest).

PANDAS & NUMPY For data manipulation, preprocessing, and numerical computations.

Technology Stack

Model Deployment

STREAMLIT — Enables fast development of interactive web apps to showcase ML models.

Frontend Technologies

HTML, CSS, JAVASCRIPT — Used for designing basic user interfaces (if required).

Supporting Tools & Platforms

JUPYTER NOTEBOOK — For exploratory data analysis and visualization

PANDAS & NUMPY — Provides cloud-based notebooks with free GPU/TPU access.

GIT & GITHUB — For version control and collaborative development.

Dataset & Input Features

Dataset Overview

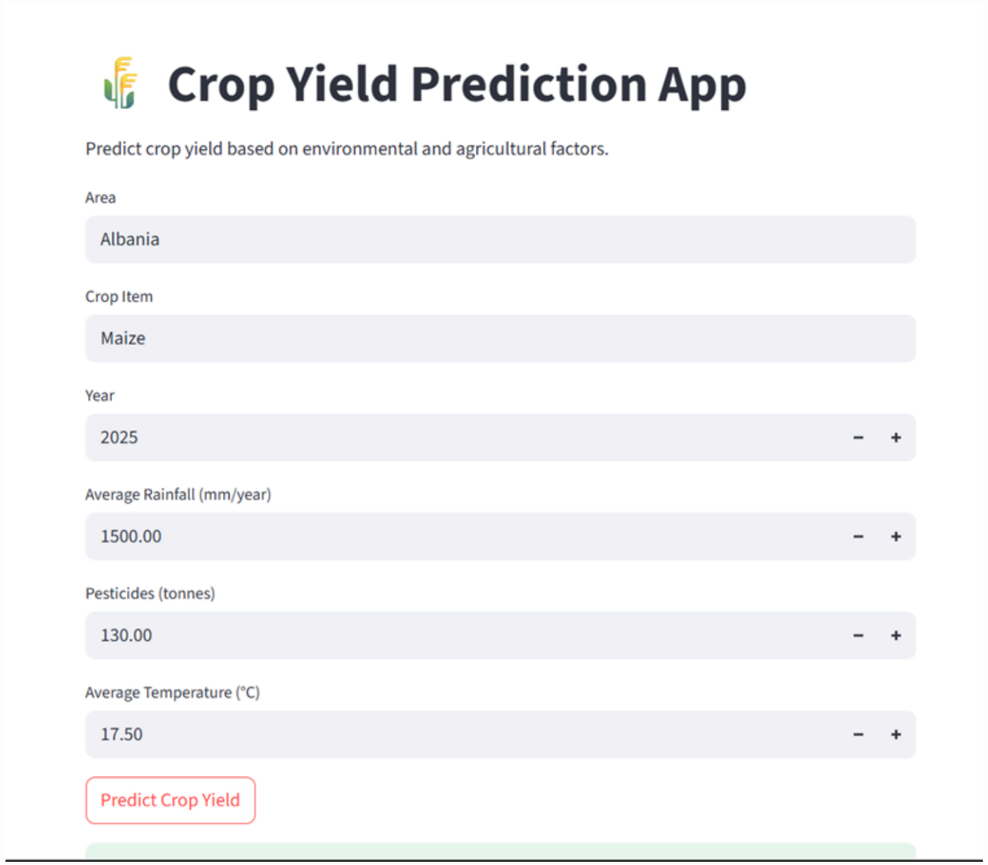
- The dataset comprises **28,242** entries and **7** key variables, collected across multiple countries and years.

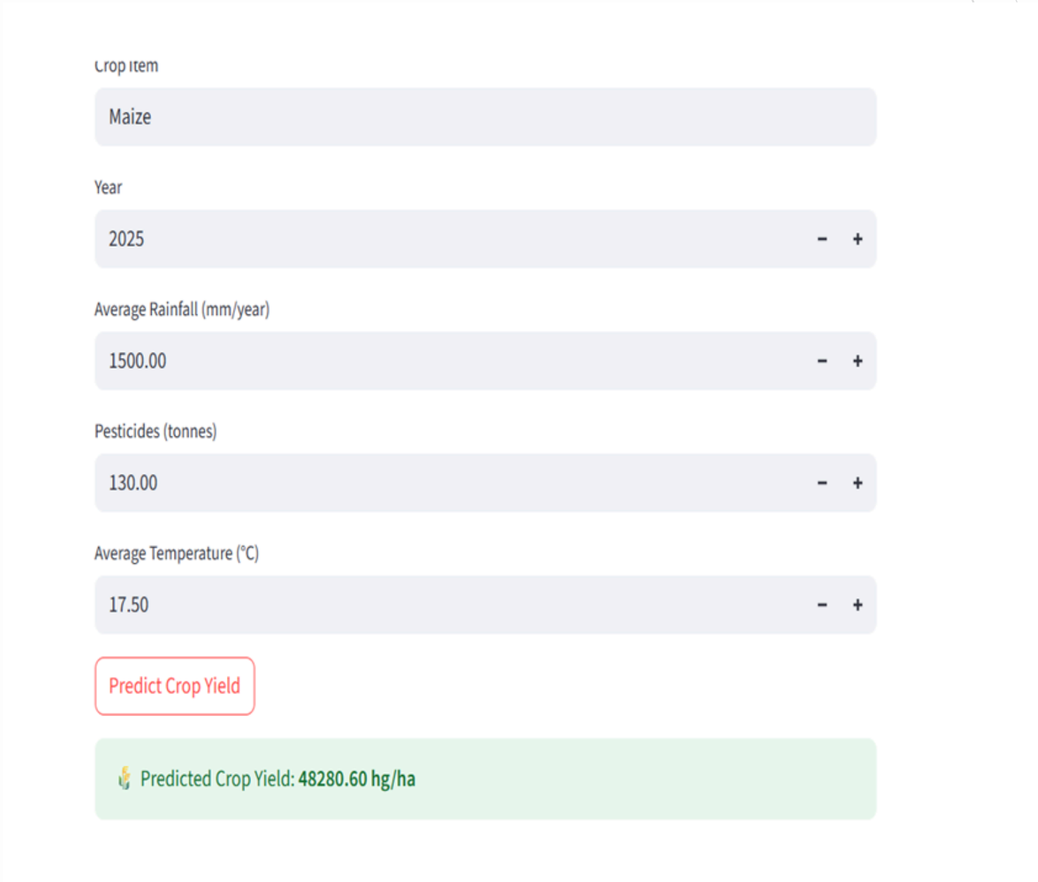
- It combines agricultural outputs with climatic and environmental conditions, providing a strong foundation for crop yield prediction using AI models.

Features Used as Inputs

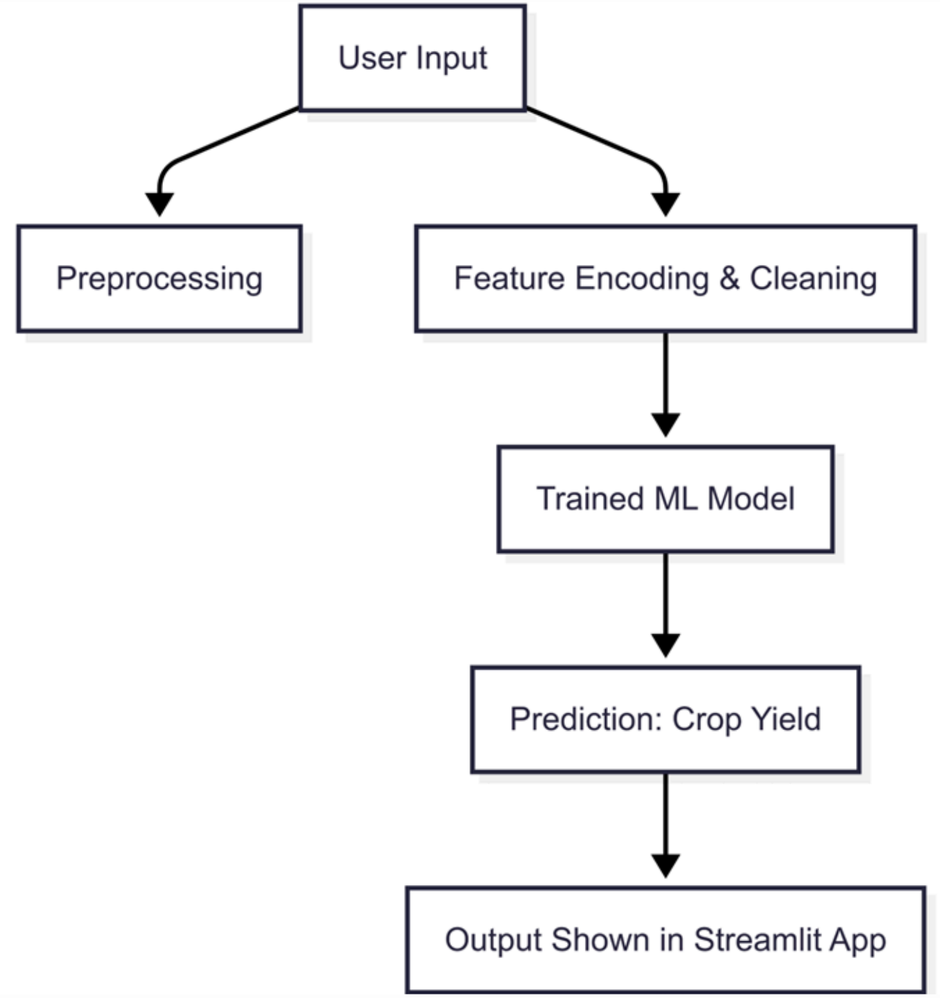
- **Area** – Geographic region or country of cultivation
- **Crop Type (Item)** – The specific crop grown (e.g., maize, rice, potatoes)
- **Year** – The year the data was recorded
- **Average Annual Rainfall** – Measured in millimeters
- **Pesticide Usage** – Amount of pesticides applied (in tonnes)
- **Average Temperature** – Average yearly temperature in °C

Web App Demo

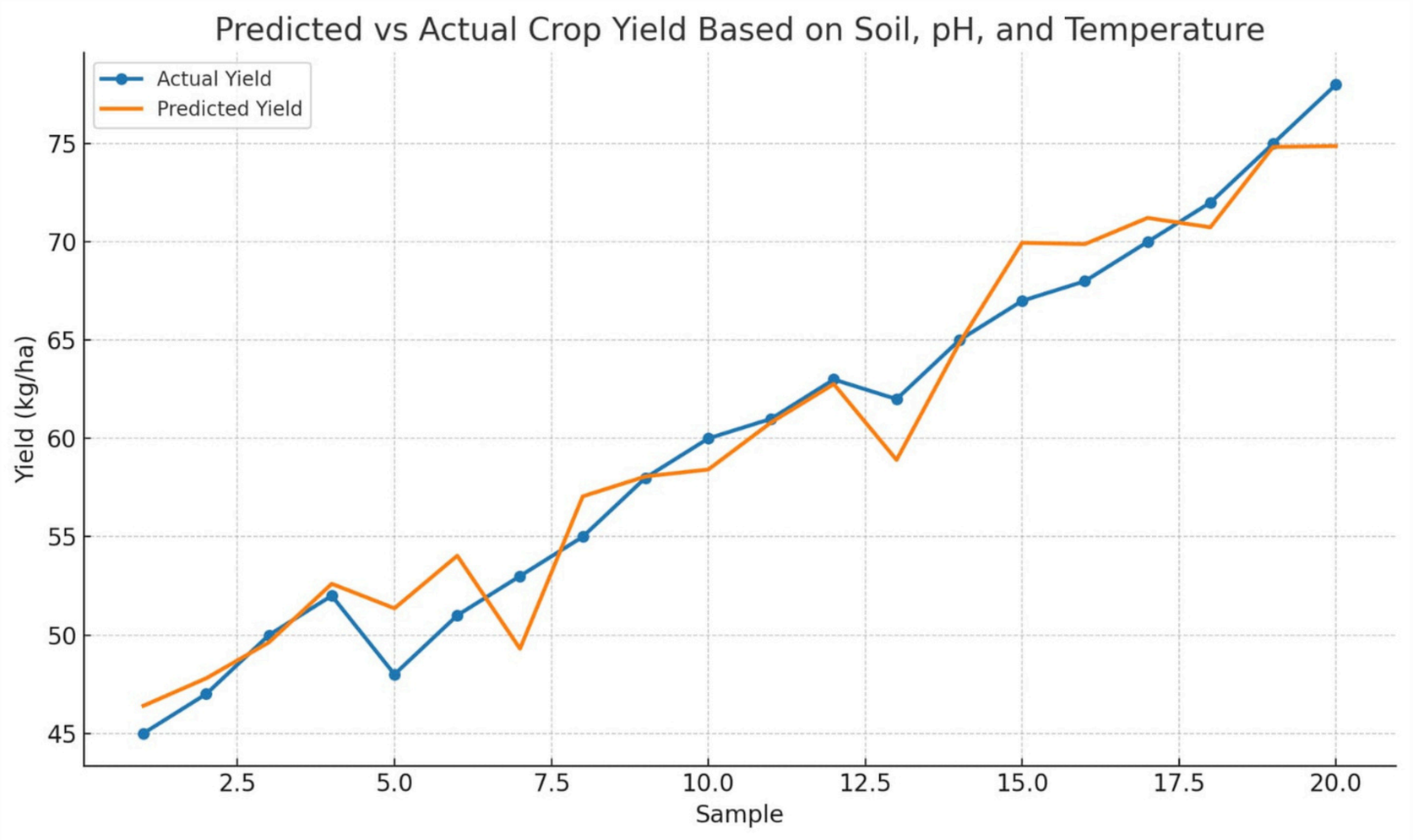

Crop Yield Prediction App
Predict crop yield based on environmental and agricultural factors.
Area
Albania
Crop Item
Maize
Year
2025
Average Rainfall (mm/year)
1500.00
Pesticides (tonnes)
130.00
Average Temperature (°C)
17.50
Predict Crop Yield


Crop Item
Maize
Year
2025
Average Rainfall (mm/year)
1500.00
Pesticides (tonnes)
130.00
Average Temperature (°C)
17.50
Predict Crop Yield
Predicted Crop Yield: 48280.60 hg/ha

Model Overflow



Graphical Analysis



Future Scope

- **Real-time Data Integration** – Use live weather and sensor data.
- **Add More Features** – Include soil quality, fertilizer use, etc.
- **Mobile App** – Build a user-friendly app for farmers.
- **Geospatial Visualization** – Show yield predictions on maps.
- **Model Enhancement** – Use advanced algorithms for higher accuracy.

Conclusion

- This project demonstrates the effectiveness of *machine learning* in predicting crop yield using key environmental and agricultural factors.

- With accurate predictions, farmers and policymakers can make informed decisions, leading to better crop planning and food security.

- The use of tools like ***Python***, ***Scikit-learn***, and ***Streamlit*** ensures a scalable and easy-to-deploy solution.

Project / Github Link

🔗 **GitHub Link:** https://github.com/Ananya92006/Crop-prediction-ibm

🔗 **Project Link:** https://ggl.link/dub

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